Jinkyeong Choi

NLP & LLM Researcher

FNAI Lab

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Research Summary

NLP and LLM researcher focused on reasoning, alignment, and agents. I am first author of Findings of EMNLP 2026 work on knowledge-graph-grounded reasoning data and fine-grained preference optimization, and WSDM 2026 work on diffusion-based sequential recommendation. My current research extends this direction to reinforcement learning for LLM agents through control-sufficient textual state representations.

Research Interests: NLP and Symbolic Reasoning; LLM Training, Alignment, and Reinforcement Learning; LLM Agents

Technical Skills: Python; PyTorch; Hugging Face Transformers; LoRA/PEFT; SQL; LLM Training and Evaluation

Education

2025.03–Present **M.S., Department of Artificial Intelligence and Data Science** — Sejong University
GPA: 4.5 / 4.5

2020.02 **B.S., Department of Computer Software Engineering** — DongEui University
GPA: 3.03 / 4.5

International Conference Publications

2026.08 **KARMA: Knowledge Graph-based Automated Reasoning Materialization and Alignment** — Findings of EMNLP 2026
NLP Preference Optimization
First author | **Accepted, forthcoming** | 2,533 of 17,669 submissions (14.3%)
▹ arXiv:2607.03166
Research Focus: Developed a knowledge-graph-grounded pipeline for structured reasoning-data synthesis and slot-level preference optimization for precise alignment.

2026.02 **SimDiffRec: Semantic Similarity-Guided Diffusion for Contrastive Sequential Recommendation** — WSDM 2026
Recommender Systems Diffusion Models
First author | **Published**, pp. 111–121 | 16.3% acceptance rate
▹ DOI: 10.1145/3773966.3777978 ▹ Code: github.com/zingyon/SimDiffRec
Research Focus: Designed semantic-similarity-guided diffusion to reduce context distortion in sequential recommendation and stabilize training.

Journal Publication

2026.07 **Sequential Recommendation via Diffusion Guided by Item Repurchase Cycles** — KIISE Transactions on Computing Practices
Recommender Systems Diffusion Models
Second author | **Published**, KCI-indexed journal, Vol. 32, No. 7, pp. 321–325
▹ KCI: ART003361598
Research Focus: Developed repurchase-aware attention that injects item-specific repurchase cycles into diffusion-based sequential recommendation.

Research in Progress

2026.08 **Learning the Rate but Not the Days: Diagnosing and Repairing Time Series Foundation Models on Zero-Inflated E-Commerce Demand** — ESWA (Under Review)
Time-Series Forecasting Foundation Models
Second author | International journal manuscript under review
Research Focus: Diagnosed zero-inflated forecasting failures and repaired them through objective and scaler corrections; SBA-ZI outperformed time-series foundation models on CRPS.

Research in Progress

2026.08–Present **Control-Sufficient Textual State Abstraction for Reinforcement Learning with LLM Agents**

LLM Agents Reinforcement Learning

Lead researcher | Work in progress

Research Focus: Developing textual state representations that preserve decision-relevant information for long-horizon LLM agent learning.

Korean Conference Publications

2026.07 **Mitigating Distribution Mismatch in Zero-Inflated Time Series Foundation Models: Mixture of Experts-Based Domain Adaptation and Log Transformation Strategies** KCC 2026

Time-Series Forecasting Domain Adaptation

Second author | Published in the KCC 2026 proceedings

2026.02 **An Efficient Travel Itinerary Generation Agent Based on LLMs with Plugin Tools and ReAct Orchestration** KICS 2026

LLM Agents Tool Use

Co-author | Published in the KICS Winter Conference 2026 proceedings

2025.07 **Diffusion Model-Based Sequential Recommendation System Leveraging Item Repurchase Cycles** KCC 2025

Recommender Systems Diffusion Models

Second author | Published in the KCC 2025 proceedings

Recognition: Best Presentation Paper Award, selected among 61 of approximately 900 papers.

2024.10 **Enhancing Structured Data Prediction Performance Using Natural Language and Numerical Embeddings** HCLT 2024

NLP Representation Learning

First author | Presented at HCLT 2024

Research & Industry Experience

2025.08–2025.12 **R&D Project Researcher** ETRI Joint R&D

Project: Generative AI model development and generalization analysis for food-sauce quality prediction.

2025.03–2026.02 **R&D Project Researcher** Yanolja Joint R&D

Project: Tourism destination recommendation through destination-connectivity analysis.

2021.06–2023.10 **Quality Assurance & AI Tester** Wisenut

Search & Chatbot QA: Designed test cases and conducted API, UI, and regression testing.

Automation & Performance: Built test-automation scripts and supported throughput and latency evaluation.

2020.09–2020.12 **Public Data Intern, National Informatization Program** NIA

Contribution: Supported a national informatization initiative by validating public-data and Open API integrity against national quality standards.

Honors & Awards

2026 **AI Seoul Tech M.S. Research Grant Recipient**, Seoul Future Foundation

2026.02 **ACM-W Scholarship Travel Grant Recipient**, WSDM 2026, sponsored by Google

2025.12 **Excellence Award**, Multicultural Research Team Best Practice Contest, KIRD

2025.07 **Best Presentation Paper Award**, KCC 2025 AI Applications Track